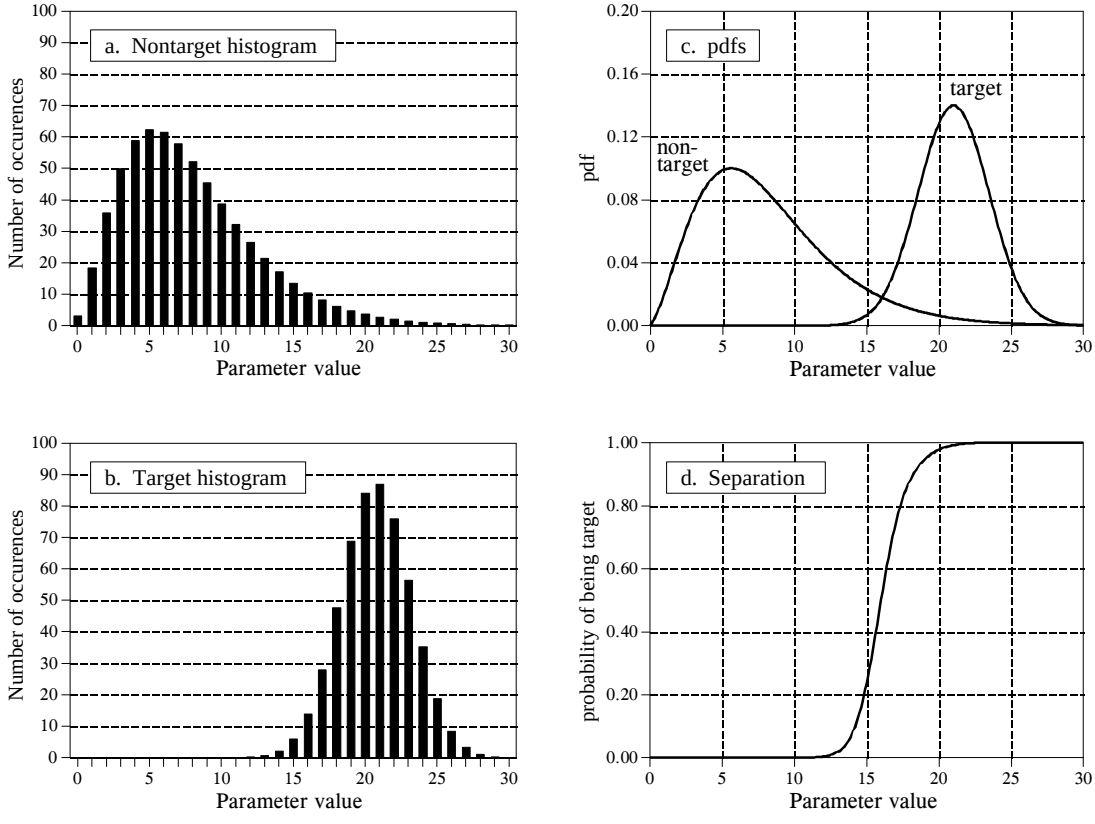

FIGURE 26-1
Probability of target detection. Figures (a) and (b) shows histograms of target and nontarget groups with respect to some parameter value. From these histograms, the probability distribution functions of the two groups can be estimated, as shown in (c). Using only this information, the curve in (d) can be calculated, giving the probability that a target has been found, based on a specific value of the parameter.

the presence of a target, since *yes* will result in one action and *no* will result in another. This is done by comparing the output value of the test to a **threshold**. If the output is above the threshold, the test is said to be **positive**, indicating that the target is present. If the output is below the threshold, the test is said to be **negative**, indicating that the target is not present. In our cancer example, a negative test result means that the patient is told they are healthy, and sent home. When the test result is positive, additional tests will be performed, such as obtaining a sample of the tissue by insertion of a biopsy needle.

Since the target and nontarget distributions overlap, some test results will not be correct. That is, some patients sent home will actually have cancer, and some patients sent for additional tests will be healthy. In the jargon of target detection, a *correct* classification is called **true**, while an *incorrect* classification is called **false**. For example, if a patient has cancer, and the test properly detects the condition, it is said to be a **true-positive**. Likewise, if a patient does not have cancer, and the test indicates that